Chocolate Milk

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Chapter

Traditional Beverages in Different Countries: Milk-Based Beverages

7.2.10 Cocoa or Chocolate Milk

Cocoa or chocolate milk drinks are dairy-based products, which are manufactured from heat-treated milk with the addition of cocoa powder and sugar and eventually stabilizers. After adding the powders, the product is heat treated with an accepted process, cooled, and filled. Apart from cocoa-based milk drink, other drinks are prepared using fruit preparations, flavors, and essences. However, some characteristics of the fruit preparation must be taken into consideration.

Chocolate milk beverage is a popular and regularly consumed food, with wide sensory acceptance among consumers of different age groups and socioeconomic levels. The product is consumed at different temperatures, and its sweet taste is an attribute that contributes to product acceptance by consumers (Shepard et al., 2013). A recent report suggests that 87% of consumers regularly purchase chocolate milk (Paixão et al., 2014). Several formulations of chocolate powder are presently available commercially, and all have a basic composition of approximately 30% sucrose or other sugars (Varnam and Sutherland, 1997). Because this fraction corresponds to a significant component of total caloric content, it presents an opportunity for the development of products with reduced levels of sugar and calories.

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2019, Milk-Based Beverages

Ali Mohamadi Sani, ... Mahya Sheikhzadeh

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ENZYMES EXOGENOUS TO MILK IN DAIRY TECHNOLOGY | Beta--Galactosidase

Flavoured Milk

Flavoured milk, including the popular chocolate milk, is usually sweetened with sucrose. This can be partially replaced by LLM giving the same sweetness but with 10% fewer calories.

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2002, Encyclopedia of Dairy Sciences

R.R. Mahoney

Chapter

Volume 3

Chocolate

In chocolate, milk fat is blended with cocoa butter for flavour and other desirable sensory qualities. When HMF is added to cocoa butter, larger crystals are formed which results in reduced crystalline interactions and a less compact network, therefore preventing extensive hardening of the product. Furthermore, addition of milk fat TAGs was found to help prevent bloom formation by delaying the transition of the β-V polymorph to β-VI polymorph which is responsible for the formation of white film on the surface of chocolate (Sonwai and Rousseau, 2010).

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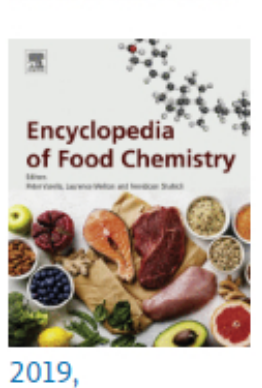


2019, Encyclopedia of Food Chemistry

P.R. Ramel, A.G. Marangoni

Chapter

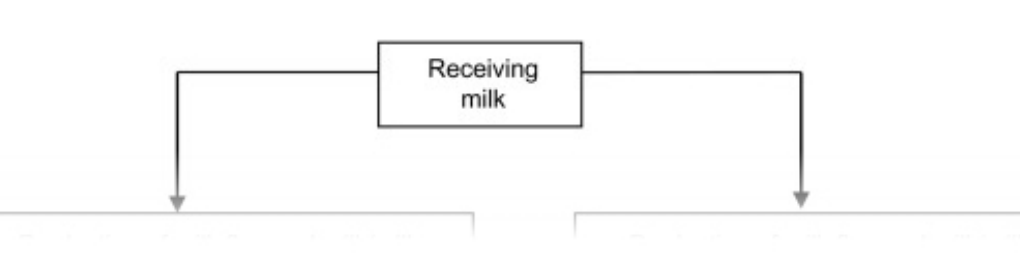
Engineering of Milk-Based Beverages: Current Status, Developments, and Consumer Trends

1.2.1 Unfermented Milk and Milk Derivatives

1.2.1.1 Flavored Milks/Drinks

Flavored milk/drink is a dairy product which has some flavors *such as* chocolate, cacao powder, fruits, and sugar, often enriched with vitamins and calcium. There is a difference between "flavored milk drink" and "the flavored milk" is owing to their fat contents. While "the flavored milk" term is used for the product which contained a milk fat percentage at least equal to the minimum legal requirement for market milk, but when the fat level is low (1%–2%), "The milk drink" term is used. Flavored milk or milk drinks have mainly been aimed at children. In several countries such as the United States and India, school meal plans contain flavored milks/drinks due to their high nutritional values. Recently, adult-friendly flavors and new packaging design has been discussed for these products in order to broaden the consumer base (Anonymous, 2017a,b,c).

Pasteurization or ultra high-temperature (UHT) sterilization process is used in the production of flavored milk and milk drinks. When the sterilization process is used, the shelf life of the product becomes long and can be stored at room temperature (Fig. 1.1). Unlike the pasteurization process, flavored milk and milk drinks can be stored at room temperature after UHT sterilization because the filling of flavored milk or milk drink in sterilized special packaging material or container (laminated cartoon or plastic bottle) is achieved in an aseptic condition. This process prevents recontamination of the products during storage and transport (Anonymous, 2017a,b,c).



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2019, Milk-Based Beverages

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Chapter

Sensory Properties of Bakery and Confectionary Products

Sensory Properties of Chocolate

Chocolate originated approximately 4000years ago at the boundary of North and Central America in the regions that now correspond to Mexico and Guatemala. This early version of chocolate was highly regarded by the local population and consumed as a bitter drink and was very different to what we describe as chocolate today. Chocolate is made from a combination of cocoa solids, sugar, cocoa butter, lecithin and, in the case of milk chocolate, milk solids (e.g., milk powder or condensed milk). In some countries, other vegetable fats may also be permitted. According to the European regulation (EC 200/36) vegetable fats other than cocoa butter, up to a maximum of 5%, are permitted in certain Member States. Additionally, milk chocolate must contain not less than 30% total dry cocoa solids and not less than 18% dry milk solids obtained by partly or wholly dehydrating whole milk, semi- or full-skimmed milk, cream, or from partly or wholly dehydrated cream, butter or milk fat, including not less than 4.5% milk fat (EC 200/36). Dark chocolate contains cocoa liquor, sugar and cocoa butter with very dark varieties containing only the butter that is present in the liquor. White chocolate contains just cocoa butter, instead of cocoa solids, as well as sugar and milk powder. There are four types of cocoa bean that originate from trees of the variety Criollo, Forastero, Trinitario and Nacional. All require high rainfall and humidity to successfully grow. Criollo is quite rare, low yielding and an expensive variety grown in Central America and the Caribbean and has a complex flavour. Forastero is the most popular and robust variety, from a flavour and cultivation perspective, which originated in the amazon but is now mostly grown in West Africa (Ivory coast, Ghana, Nigeria), but also in southern and central America as well as Sri Lanka, Malaysia and Indonesia. Trinitario originates from Trinidad and is a hybrid of Forastero and Criollo varieties. Finally, the rarest chocolate variety is Nacional, a member of the Forastero family with up to 40% white beans and once thought extinct, but was recently rediscovered in a

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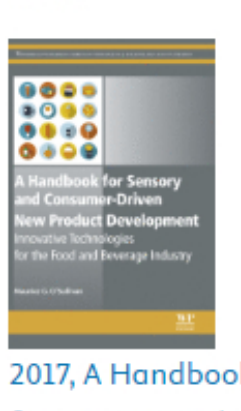

2017, A Handbook for Sensory and Consumer-Driven New Product Development

Maurice G. O'Sullivan

Chapter

Molecular gastronomy, chefs and food innovation: an interview with Michael Bom Frøst

Michael

Torsten Vildgaard, our chef, once came to us with a vision of a milk skin that he thought could contribute to making a wonderful dish. His inspiration came from several sources. Hot chocolate milk forms a thin skin on the surface, and Torsten had always wanted to use a skin of this type in a dish. Further, his inspiration was from Japanese cuisine, a product called Yuba made from soy milk. When you heat soy milk then you can peel off a thin skin from the top. The milk skin had been used at El Bulli in Spain and in one of their cookbooks they supply a recipe for it, but we were unsuccessful in making a milk skin from it using cow's milk.

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2010, Consumer-Driven Innovation in Food and Personal Care Products

M. Bom Frøst, S.R. Jaeger

Chapter

Volume 5

Chocolate Milk and Cocoa Powder

A main ingredient of chocolate milk is cocoa powder; sometimes, chocolate powder is also added. Cocoa powder used in chocolate milk is usually produced from defatted cocoa, from which cocoa butter has already been separated by pressing. The cocoa powder thus prepared has a fat content of about 11%. The particle size of the cocoa powder used in chocolate milk has to be as small as possible; 10–30μm is recommended, and ideally less than 0.5% of particles present should have a particle size above 75μm. According to Stokes' law, in addition to the increased viscosity of the liquid phase, this is an important factor in determining the rate of sedimentation (Anonymous, 2003; Boomgaard etal., 1987).

The chosen cocoa powder must also provide a good chocolate flavor and color. Alkalized types of cocoa powder are generally preferred; the more alkalized the powder, the darker the color and the stronger the flavor obtained. The strength of interactions between the casein micelles and cocoa particles increases at higher levels of alkalization.

Cocoa powder has to be free from enzymes such as heat-stable lipases and proteases and have a shell content of<1.5%. The higher the cocoa powder content, the lower the heat stability. Cocoa powder can contain heat-resistant spores surviving UHT-treatment; therefore, bacterial count including spores of cocoa powder for chocolate milk has to be low. Cocoa powder contains an insoluble cell structure, only 30%–40% of the cocoa powder is soluble, and the rest has to be kept in suspension by the thickener or gelling agent added, as the density of these particles is higher than that of milk. Usually, an indication is given on the container to shake the beverage well before use, to redistribute sedimented cocoa particles (Anonymous, 2003). Cell walls of cocoa powder may be enzymatically treated to increase solubility and reduce sedimentation (Holkar etal., 2019).

The amount of cocoa and/or chocolate powder added varies between 1% and 10%, depending on the quality and price range of the product. Other common ingredients for chocolate milk are sucrose (often 5%–9%) or other sugars or sweeteners, and often vanillin or vanilla extract.

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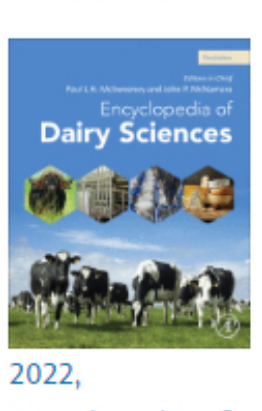


2022, Encyclopedia of Dairy Sciences (Third Edition)

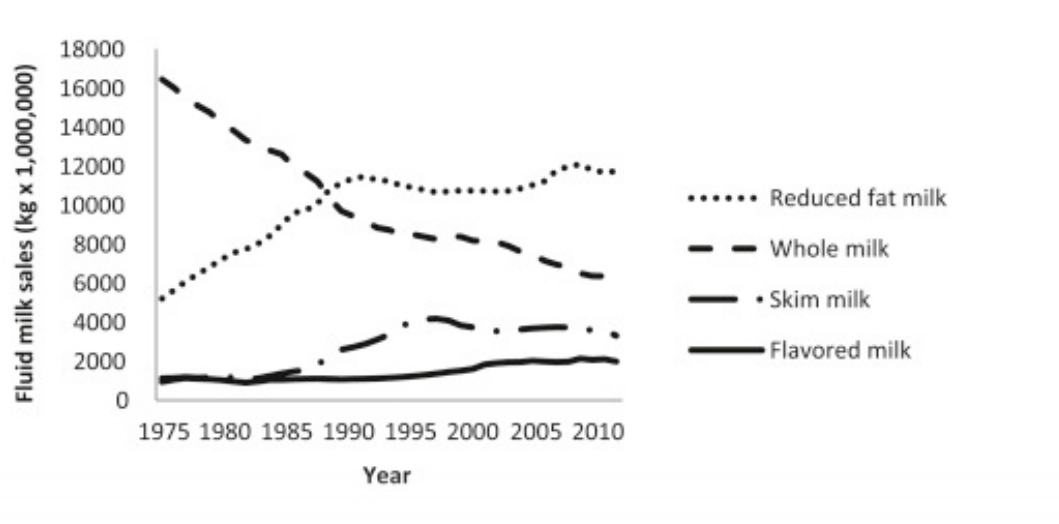
Walter Bisig, Alan L. Kelly

Chapter

Liquid Milk Products: Pasteurized Milk

Types of Pasteurized Milks

Thirty years ago, the milk beverage of choice in North America was whole milk, which contains a minimum of 3.25% fat. Consumer concerns over high-fat diets have led to a shift in milk consumption away from whole milk toward reduced-fat, low-fat, or skim milk (Figure2). The nutritional quality of low-fat milk is similar to that of whole milk, although there is a decrease in fat-soluble vitamins and in caloric value. Skim milk has different sensory characteristics (whiteness, mouthfeel, flavor) than whole milk. Therefore, many skim milks have been developed with extra solids content, water-binding stabilizers, and other ingredients to imitate the texture and mouthfeel of whole milk. Vitamin D-fortified milk was introduced to the United States in 1933 to enhance calcium absorption and reduce rickets and it is common to fortify reduced- and low-fat milks with both vitamins A and D. Vit A addition is required while Vit D addition is voluntary. Basically, these fat-soluble vitamins are added back to make up for vitamins removed in the cream. Vitamins A and D are added to obtain levels of at least 2000 and 400 international units (IU) per quart (~0.951) milk, respectively.



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2016, Reference Module in Food Science

Lisbeth Meunier-Goddik, S. Sandra

Chapter

LABELING OF DAIRY PRODUCTS

Application to milk products

The name 'milk'

The term 'milk' is defined as the normal, untreated (i.e., raw) mammary secretion of any milking animal obtained from one or more milkings without either addition to it or extraction from it.

The Codex Standard for the Use of Dairy Terms permits the use of the term milk for treated raw milk; however, if the milk is modified or adjusted in composition, the term may only be used as a name if a clear description of the modification made is given in close proximity to it, for instance, 'skim milk', 'lactose-reduced milk', and 'vitaminized milk'.

Naming of milk products

Codex has established names for the milk products of significance in international trade. These products shall be named as specified in the relevant standard, and should comply with the content of it. However, the use of defined variety names of cheeses (e.g., Cheddar, Emmental, and Brie) and of fermented milks (e.g., yogurt and kefir) is optional – the variety name can replace the product designation if the product complies with the corresponding standard(s). Those milk products to which a name has not been established by Codex shall be named as specified in the national legislation in the country of sale, or by appropriate descriptive designations in the absence of a prescribed name. Milk products not covered by Codex commodity standards are listed in **Table 3**.

Table 3. Milk products not covered by Codex commodity standards

Preserved milk products	**Fermented milk powders, e.g., yogurt powder and buttermilk powder**
	Whey protein concentrates and other driedindividual milk constituents

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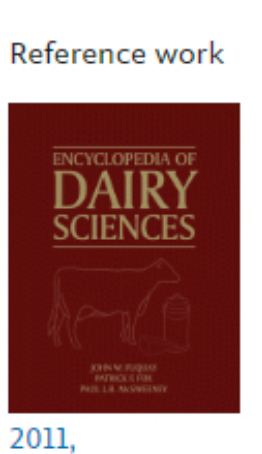

2011, Encyclopedia of Dairy Sciences (Second Edition)

C. Heggum

Chapter

Liquid Milk Products: Flavored Milks

Processing of Chocolate Milk

For a good swelling of the cocoa particles and therefore an improved dispersability, the cocoa particles are first mixed into milk at a ratio of 1:2 and held for 2–3h. Sometimes, this mixture is heated up to 80–90°C for 30min instead. This also allows air incorporated in the cocoa particles to escape. Carrageenan is premixed with three to five parts of sugar before being added to the milk, considering its tendency to form lumps. The concentrated suspension of cocoa in milk, the premixed carrageenan, and the other ingredients are mixed into the main part of the milk at 40°C. Intense mixing keeps the cocoa particles in suspension during manufacture (Anonymous, 2003).

For full-fat and semi-skimmed flavored milks, homogenization is applied either before (pasteurization, ultra-pasteurization) or after the heat treatment (UHT). Homogenization at the recommended pressure of 180–200bar reduces the size of the fat globules but not the size of the cocoa particles. Heat treatment according to the conditions mentioned in article "definition of flavored milks" is followed by cooling and filling. Above 60°C, the carrageenan molecules dissolve completely and interact more with the casein micelles. Pasteurized and ultra-pasteurized products are cooled to 4°C and filled under ultra-clean conditions into HDPE plastic bottles or beverage cartons. UHT products are cooled to 20°C and filled aseptically. After reaching a temperature below 35°C, the carrageenan helices interact with neighboring helices and form a three-dimensional network leading to weak gelation. This will only happen if the flavored milk is left undisturbed. The processing equipment can be designed without the need to consider viscosity buildup. Nevertheless, once the flavored milk is cooled below the gelling temperature of carrageenan (around 35°C), it should not be pumped or stirred for long. Sterilized flavored milks are heat-treated after filling. Cooling the milk before filling to a lower temperature increases the strength of the gel. The filling temperature should not be higher than 20°C. If a higher filling temperature is used, there will not be sufficient thixotropy present to obtain full stability, and cocoa particle sedimentation may occur. Even for UHT-treated chocolate milk, cooling to 5°C is advised to prevent sedimentation, even where it is intended that storage will be at room temperature. In-container-sterilized flavored milks must be rotated during cooling to allow the forming network to entrap the cocoa particles and to keep them in suspension (Anonymous, 2003).

Overall, important quality criteria for chocolate milk include a homogeneous appearance without phase separation or too much sediment, an appealing brown color, a good authentic chocolate flavor, and clean flavor release, a viscosity high enough to give creaminess and a favorable mouthfeel but avoiding a strong gel, low counts of Enterobacteriaceae and aerobic mesophilic bacteria, a good dispersability of a possible sediment, a fat content according to definition, a pH value in the range 6.2–7.0, and stability during a storage test for the defined shelf life.

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2017, Reference Module in Food Science

Walter Bisig, Alan L. Kelly

Related terms:

Carrageenan, Enterotoxin, Alginic Acid, Dairy Product, Flavored Milk, Chocolate, Cocoa Powder, Milk Chocolate, Yogurt, Kappa Carrageenan.

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